

Lecture 3: Training and Fine-Tuning LLMs

Large Language Models in Finance

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Financial Data: A Tiny Slice of the Web

The core problem

Financial text constitutes roughly **0.5%** of a typical filtered web crawl. A model that has seen the distribution of the web will be fluent in casual prose and weak on financial contracts.

Four key financial text sources:

- ① **EDGAR** — 10-K, 10-Q, 8-K, proxy filings; full regulatory English; public from 1996 to present
- ② **Earnings-call transcripts** — spoken-register financial language, Q&A with management; available via commercial vendors or scraping
- ③ **Bloomberg news** — licensed; high-density, entity-linked financial news with multi-decade coverage
- ④ **Alternative data** — satellite-imagery metadata, credit-card transaction summaries, job-posting feeds, shipping manifests

Consequence

Low financial representation → high tokeniser fertility on financial text →

Tokenisation: BPE and Fertility

Byte-pair encoding (BPE) iteratively merges the most frequent adjacent symbol pair until the target vocabulary size is reached.

Fertility formula:

$$F(\tau, C) = \frac{\text{tokens produced by } \tau \text{ on corpus } C}{\text{whitespace-separated words in } C}$$

Fertility ≈ 1.0 means efficient coverage; fertility > 1.5 means heavy fragmentation of domain vocabulary.

Text type	Fertility (GPT-2 BPE)	Effect
Standard English prose	1.10–1.20	Efficient
Financial regulatory text	1.40–1.60	Fragmented

Words like CUSIP, EBITDA, and ISDA may be split into 3–5 sub-tokens by a general BPE tokeniser, each consuming an attention slot. Training a domain-specific tokeniser alongside domain-specific weights reduces fertility and shortens effective sequence lengths.

Data Quality Pipeline

Raw web crawl data is $\approx 90\%$ noise. A standard pipeline applies four stages:

- 1 **Language identification** — fastText classifier assigns each document a language label; non-target languages discarded
- 2 **Heuristic filters** — punctuation rate, symbol-to-word ratio, minimum token count, repeated n-gram detection
- 3 **Perplexity filtering** — small KenLM reference model scores each document; high-perplexity outliers removed
- 4 **MinHash deduplication + PII scrubbing** — near-duplicate removal by Jaccard similarity; removal of emails, phone numbers, SSNs

MinHash: Jaccard similarity definition

$$J(A, B) = \frac{|S(A) \cap S(B)|}{|S(A) \cup S(B)|}$$

where $S(d)$ is the set of k -gram shingles of document d . Documents with $J > 0.8$ are near-duplicates; all but one copy is removed.

Pre-training Objectives: CLM, MLM, Span Corruption

Objective	Loss	Best suited for
CLM (causal LM)	$-\sum_t \log P(x_t x_{<t})$	Generation, chat, decoder-only (GPT)
MLM (masked LM)	$-\sum_{t \in \mathcal{M}} \log P(x_t x_{\setminus \mathcal{M}})$	Classification, NER, encoder-only (BERT, FinBERT)
Span corruption (T5)	Reconstruct masked spans from sentinels	Seq2seq, QA, summarisation

CLM uses causal (lower-triangular) attention masking. MLM attends *bidirectionally* to all unmasked tokens. Span corruption replaces runs of consecutive tokens with a single sentinel `<extra_id_k>` and trains the decoder to produce all spans in order.

Finance rule of thumb

Chinchilla Scaling Laws: Setup

Hoffmann et al. (2022) fit the empirical loss of an autoregressive LM as:

$$L(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}$$

- N = number of trainable parameters
- D = number of training tokens
- E = irreducible entropy floor
- $\alpha, \beta > 0$ govern the decay rate of each term

Compute constraint:

$$C \approx 6ND$$

(factor 6: 2 for forward pass, 4 for backward pass and parameter update, standard mixed-precision accounting)

The question

Given a fixed FLOP budget C , what is the optimal split (N^*, D^*) ?

Chinchilla Scaling Laws: Result

Minimising $L(N, D)$ subject to $C = 6ND$ gives:

$$N^*(C) \propto C^{\beta/(\alpha+\beta)}, \quad D^*(C) \propto C^{\alpha/(\alpha+\beta)}$$

With Chinchilla estimates $\alpha \approx 0.34$, $\beta \approx 0.28$:

$$\frac{D^*}{N^*} \approx 20$$

Worked example: 7B-parameter model

	GPT-3 (175B)	Chinchilla-optimal 7B
Parameters	175B	7B
Training tokens	300B	140B
D/N ratio	1.7	20
Status	Over-parameterised	Optimal

Practical implication

Distributed Training: Three Parallelism Types

Type	What is partitioned	Key mechanism
Data parallelism	Batch (each device holds full model)	Ring-AllReduce gradient aggregation
Tensor / model parallelism	Weight matrices column-wise; $W \in \mathbb{R}^{d \times k} \rightarrow [W_1 W_2 \dots]$	All-gather / all-reduce intra-layer
Pipeline parallelism	Layer groups to successive devices	Micro-batches; bubble fraction $\approx \frac{p-1}{m+p-1}$

ZeRO (Zero Redundancy Optimizer) stages:

- **Stage 1:** partition optimiser states across devices
- **Stage 2:** also partition gradients
- **Stage 3:** also partition model weights $\rightarrow$ near-linear memory reduction with device count

Engineering Efficiency: bfloat16, Checkpointing, FlashAttention

Mixed-precision training (bfloat16):

- Weights stored in bfloat16 (16-bit); fp32 master copy for optimiser
- Halves memory and HBM bandwidth versus fp32
- Loss scaling: multiply loss by s before backward; divide gradients by s to prevent underflow

Gradient checkpointing:

$$\text{Activation memory} = O(L) \xrightarrow{\text{checkpoint every } \sqrt{L}} O(\sqrt{L})$$

Discards intermediate activations; recomputes them during backward pass.
Cost: one extra forward pass.

FlashAttention: Tiles QK^T into SRAM-resident blocks; avoids writing the full $N \times N$ attention matrix to HBM. Reduces HBM reads/writes from $O(N^2)$ to $O(N^2/M) \cdot M$ (subquadratic in practice). Exact attention, no approximation.

The Adaptation Spectrum

Method	Params dated	up-	Data needed	Best use case
Zero-shot	0		0	Well-defined tasks in training distribution
Few-shot (ICL)	0		1–100 examples	Novel formats; no gradient compute
PEFT (LoRA)	0.1–1%		1K–100K	Domain shift; resource-constrained
Full fine-tuning	100%		100K+	Max in-domain accuracy; large data budget

LoRA: Low-Rank Adaptation

For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA sets:

$$W = W_0 + BA, \quad B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times k}, \quad r \ll \min(d, k)$$

W_0 is frozen. Only B and A are trained. Trainable parameters per layer: $r(d + k)$.

Worked example: 7B LLaMA, $r = 8$, all 4 projections, 32 layers

$$32 \times 4 \times 8 \times (4096 + 4096) = 8,388,608 \approx 8.4\text{M}$$

$$\frac{8.4\text{M}}{7,000\text{M}} \approx 0.12\%$$

	LoRA ($r = 8$)	Full fine-tuning
Trainable params	8.4M (0.12%)	7,000M (100%)
A100 training time	≈ 90 min	≈ 40 hr
Inference overhead	Zero (merge BA)	None

Prefix Tuning and Adapters

Prefix tuning:

Learnable prefix vectors are prepended to K and V at every layer:

$$K' = \begin{bmatrix} P_K \\ K \end{bmatrix} \in \mathbb{R}^{(l+n) \times d_k}, \quad V' = \begin{bmatrix} P_V \\ V \end{bmatrix} \in \mathbb{R}^{(l+n) \times d_v}$$

Only the prefix matrices P_K, P_V (of length l) are trained. Effective for generation tasks; steers the output distribution without touching weights.

Adapters:

A bottleneck module inserted after each sub-layer:

$$h \leftarrow h + W_{\text{up}} \text{ReLU}(W_{\text{down}} h), \quad W_{\text{down}} \in \mathbb{R}^{d \times m}, \quad W_{\text{up}} \in \mathbb{R}^{m \times d}, \quad m \ll d$$

Adds inference latency unless fused.

Why LoRA wins in practice

Zero inference overhead (merge $W_0 + BA$ before serving); multiple adapters swappable for different tasks; QLoRA extends to 4-bit quantised base models.

RLHF: Three-Stage Alignment Pipeline

Stage 1 — Supervised Fine-Tuning (SFT): Train base model on high-quality human-written demonstrations. Produces initial policy π_{SFT} .

Stage 2 — Reward Model Training: Human annotators rank pairs of responses ($y_w \succ y_l$) for the same prompt x . Bradley-Terry loss:

$$\mathcal{L}_{\text{RM}} = -\mathbb{E}[\log \sigma(r_\theta(x, y_w) - r_\theta(x, y_l))]$$

Stage 3 — PPO fine-tuning with KL penalty:

$$\mathcal{L}_{\text{RL}}(\theta) = \mathbb{E}_{\pi_\theta}[r_\phi(x, y) - \beta \text{KL}(\pi_\theta(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x))]$$

β controls how far the policy can drift from the SFT reference.

Limitations for finance

Reward models can be gamed; PPO is unstable; human annotation is expensive. A single annotator disagreement on “plausible vs. accurate” in a financial context can propagate to thousands of training examples.

Direct Preference Optimisation (DPO)

DPO eliminates the reward model entirely. It directly optimises the policy on preference pairs by re-parameterising the reward in terms of log-ratios:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

Key properties:

- No separate reward model to train or maintain
- Same preference data as RLHF; simpler single-stage training loop
- Empirically competitive with RLHF on alignment benchmarks
- β still controls implicit KL penalty vs. π_{ref}

Interpretation

DPO increases the log-probability of the preferred response y_w relative to π_{ref} while decreasing the log-probability of y_l — a direct implementation of the preference signal.

FinBERT: Domain-Adapted Encoder for Finance

Architecture: BERT-large continued pre-training on $\approx 4.9\text{B}$ tokens of Reuters news, Financial PhraseBank, and analyst reports (MLM objective).

Classification head:

$$\hat{y} = \text{softmax}(W_c \cdot h_{[\text{CLS}]} + b_c)$$

$h_{[\text{CLS}]}$ is the final-layer representation of the [CLS] token.

Empirical results on Financial PhraseBank (all-agree split):

Model	Accuracy
Human upper bound	$\approx 97\%$
FinBERT	88.5%
BERT-large (general)	80.7%
Dictionary / rule-based	$\approx 72\%$

The 7.8 pp gain over BERT-large comes almost entirely from domain-specific continued pre-training, not architectural changes.

BloombergGPT: A 50B Finance Decoder

Training corpus: 363B tokens total — 51% Bloomberg financial data (news, filings, transcripts), 49% general web and books.

ALiBi positional encoding:

$$\text{score}(q_i, k_j) = \frac{q_i^\top k_j}{\sqrt{d_k}} - m(i - j)$$

where m is a head-specific slope and $(i - j)$ is query-key distance. ALiBi extrapolates more gracefully to sequences longer than those seen in training compared to learned absolute positional embeddings.

Chinchilla lens on BloombergGPT:

	BloombergGPT	Chinchilla optimal
Parameters	50B	50B
Training tokens	363B	$\approx 1\text{T} (= 50\text{B} \times 20)$
Status	Under-trained by Chinchilla criterion	

Lesson: even a well-resourced organisation trained a model that violated

Evaluation: Perplexity, Macro-F1, ROUGE

Perplexity (language modelling):

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(x_t \mid x_{<t})$$

Three limitations: not a task metric; gameable by spreading mass; requires token-level probabilities.

Macro-F1 (classification):

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K \frac{2 P_k R_k}{P_k + R_k}$$

Treats all classes equally regardless of frequency. Appropriate for imbalanced financial datasets (rare event classes matter as much as common ones).

ROUGE-N (summarisation):

$$\sum \sum \text{count}_{\text{match}}(g)$$

Red-Teaming: Financial Failure Modes

Four recurring failure modes identified by red-teaming:

- ① **Factual hallucination** — generates plausible but incorrect price forecasts or earnings figures
- ② **Temporal hallucination** — cites outdated regulations as current (e.g., pre-2018 MiFID II rules)
- ③ **Regulatory compliance failure** — provides what appears to be personalised investment advice without required licensing disclosures
- ④ **Herd-bias amplification** — echoes analyst consensus while suppressing dissenting signals

Empirical illustration: Lopez-Lira & Tang (2023)

A long-short strategy based on ChatGPT sentiment scores achieved a Sharpe ratio > 6 in 2021, declining to ≈ 1 by late 2024 as the signal became widely known and arbitrated away. *Any published alpha source degrades once it is public.*

Four Bias Types in Financial LLMs

Survivorship bias

Training corpora overrepresent successful firms: public companies file EDGAR; bankrupt or never-public firms do not. The model learns a rosier world than the one practitioners inhabit.

Temporal bias

Models trained on historical data are deployed in structurally different market regimes (post-pandemic, rising rates, de-globalisation). Performance observed on historical test sets may not persist.

Geographic bias

English-language, US-centric text dominates open web corpora. Non-US financial conventions, currencies, and legal structures are underrepresented.

Analyst consensus bias

Hallucination Types and Mitigations

Two hallucination types:

- **Factual hallucination** — fabricated prices, dates, financial figures not present in any training document
- **Regulatory hallucination** — describes rules that are outdated, misquoted, or jurisdiction-mismatched

Four principal mitigations:

- ① **Retrieval-Augmented Generation (RAG)** — ground outputs in retrieved, verified passages; reduces reliance on parametric memory
- ② **Confidence calibration** — train model to express uncertainty; reject or flag responses below a calibrated confidence threshold
- ③ **Output verification** — secondary model or rule-based checker flags claims inconsistent with a structured knowledge base
- ④ **Human-in-the-loop** — route low-confidence or high-stakes outputs to expert review before delivery to end users

No single mitigation is sufficient

Production deployments should layer at least two mitigations, with audit

MiFID II Article 37:

- Investment research must be independent from commercial pressures
- LLM contributions require demonstrable *suitability* (analysis matches recipient risk profile) and *explainability*
- Attention weight visualisations are *not sufficient* explanations: they reflect co-occurrence patterns, not causal reasoning chains

SR 11-7 (Federal Reserve Model Risk Management):

- Tiered validation proportional to model risk
- High-risk models (credit, capital, trading) require independent validation, assumption documentation, periodic performance monitoring

EU AI Act — August 2026 deadline for high-risk systems:

- Conformity assessments, technical documentation, human oversight
- Registration in EU database for high-risk financial AI applications
- High-risk: models influencing credit scoring, insurance pricing, investment decisions at scale

Governance: Risk Classification and Four-Component Framework

Three-tier risk classification:

Tier	Examples	Controls required
Low	Internal Q&A, document search	Standard access controls, logging
Medium	Earnings summarisation, first-pass compliance screening	Audit trails, human review sample
High	Credit decisioning, trade recommendation, regulatory filing	Full SR 11-7 validation, EU AI Act registration

Four-component governance framework:

- 1 **Risk classification** — assign Low/Medium/High to each use case based on decision autonomy and downstream impact
- 2 **Audit trails** — capture input prompts, retrieved context, model

Lecture 3 Wrap-Up: Key Takeaways

Data: Financial text is 0.5% of the web. EDGAR, transcripts, and Bloomberg are the primary sources. Quality pipelines and MinHash deduplication are essential.

Scaling: Chinchilla says $D^*/N^* \approx 20$. GPT-3 was over-parameterised; BloombergGPT was under-trained. Size alone is not the answer.

Fine-tuning: LoRA achieves 0.12% of 7B parameters for 90 min of training vs. 40 hr full fine-tuning, with zero inference overhead.

Alignment: RLHF (SFT $\rightarrow$ reward model $\rightarrow$ PPO) aligns behaviour. DPO simplifies it to a single supervised objective on preference pairs.

Finance: FinBERT: +7.8 pp over BERT on sentiment. BloombergGPT: under-trained per Chinchilla. FinBen: LLMs still struggle on forecasting.

Governance: MiFID II + SR 11-7 + EU AI Act 2026 require explainability, audit trails, and tiered validation. Attention weights are not explanations.

Lecture 4: AI Agents in Finance